



SwingNetV2



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목차

● 서비스 소개	-----	3
● 데이터	-----	15
● 모델 개발	-----	21
● 검증 및 테스트	-----	35
● 기대효과	-----	38

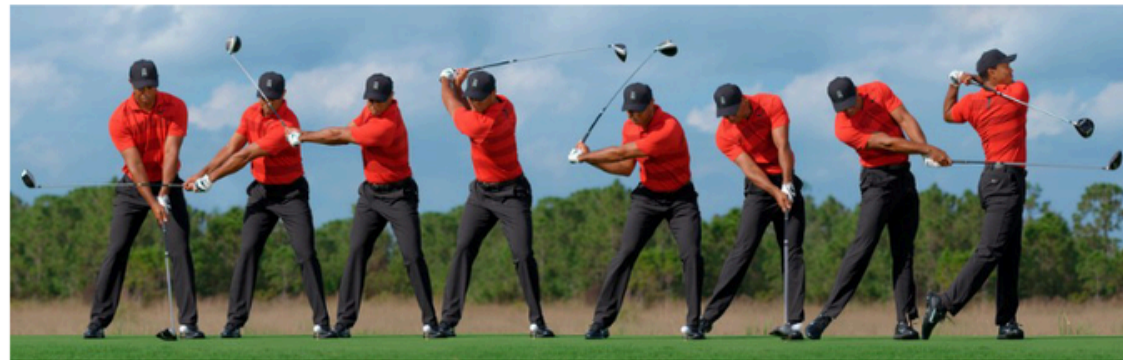
GolfDB

GolfDB

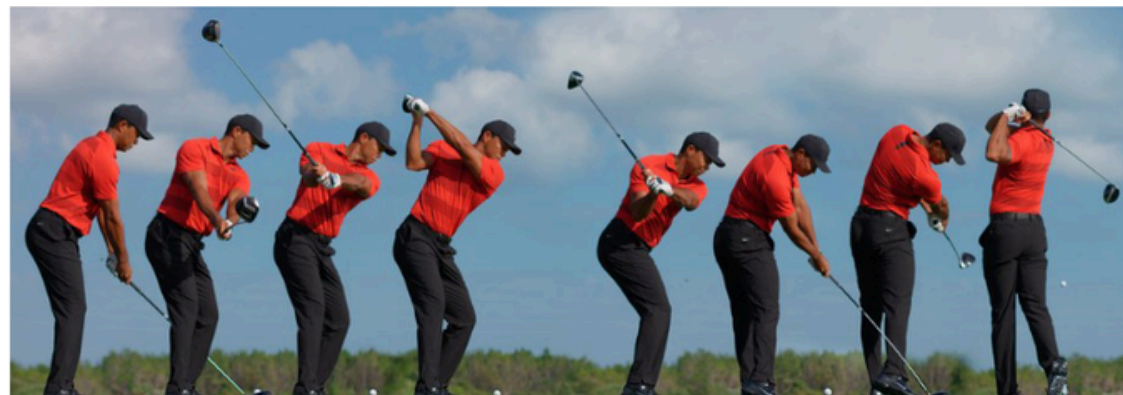
Introduced by McNally et al. in [GolfDB: A Video Database for Golf Swing Sequencing](#)

GolfDB is a high-quality video dataset created for general recognition applications in the sport of golf, and specifically for the task of golf swing sequencing.

Source: [GolfDB: A Video Database for Golf Swing Sequencing](#)



[그림1] Face-on View

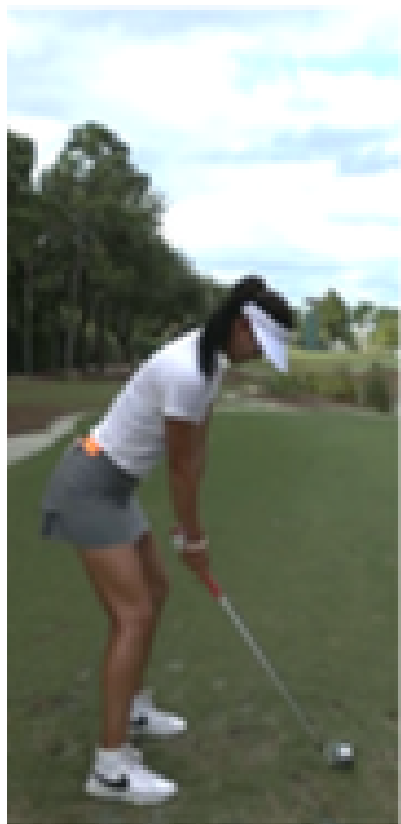


[그림2] Down-the-line View

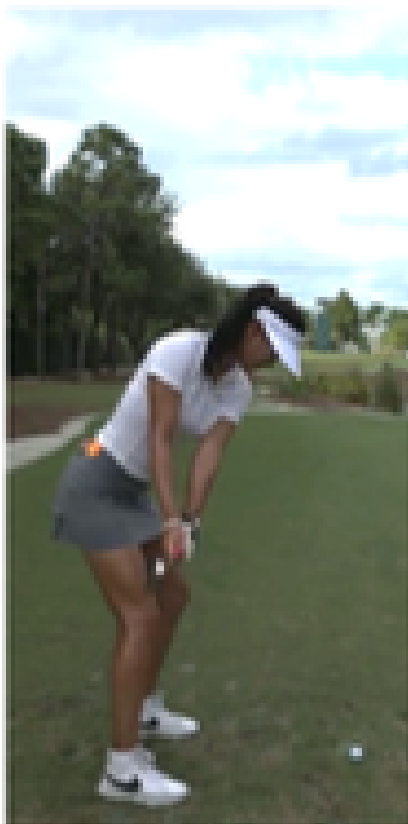
- 데이터 규모: 남녀 프로 골퍼 1,400명의 HD 골프 스윙 영상으로 구성되며, 실시간 촬영 영상과 슬로우 모션 영상이 혼합
- 총 39만 여 프레임, 8개 이벤트 라벨 제공
- 골프 스윙 시퀀스에서 8가지 주요 동작 이벤트:
Address, Toe-up, Mid-backswing, Top, Mid-downswing, Impact, Mid-follow-through, Finish
- View 타입: Face-on & Down-the-line
⇒ 다양한 각도·클럽·성별 포함
- 추가 메타 정보: fps, 클럽 종류, 성별 등
⇒ 세분화 분석·도메인 적합성 ↑

GolfDB

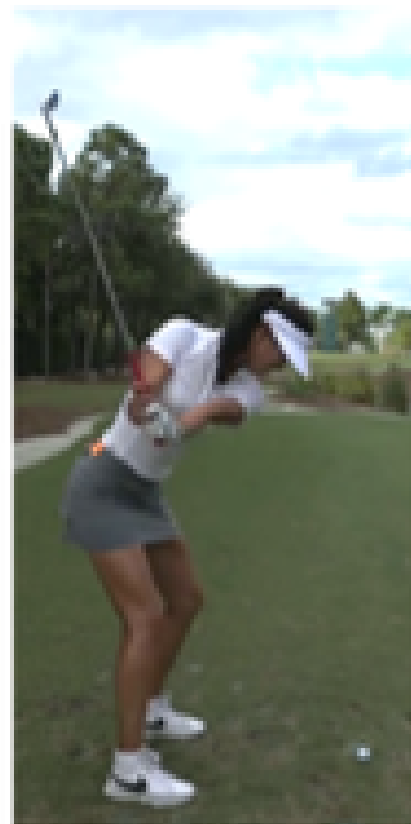
8가지 이벤트 라벨



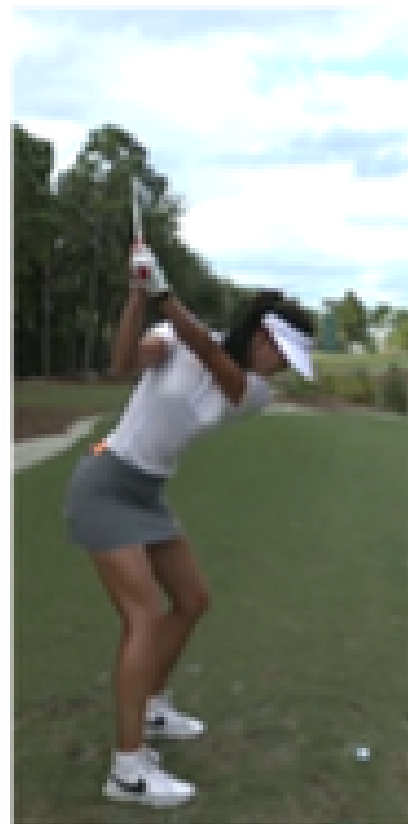
Address



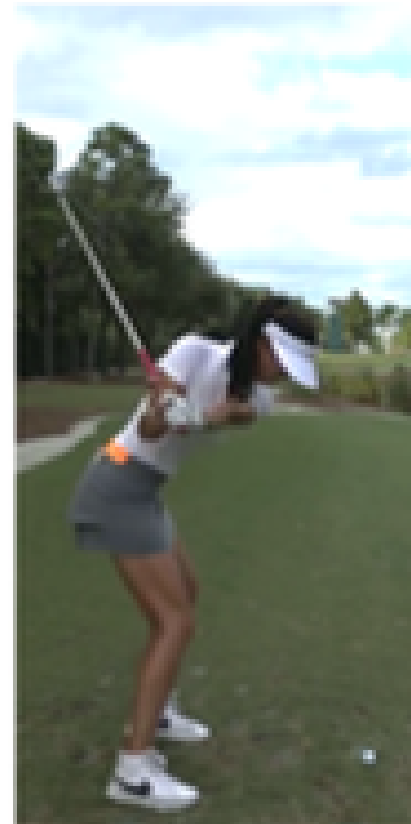
Toe-up



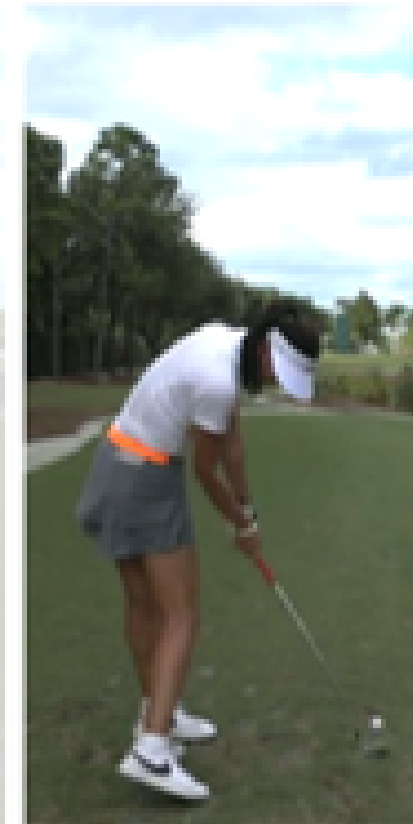
Mid-backswing



Top



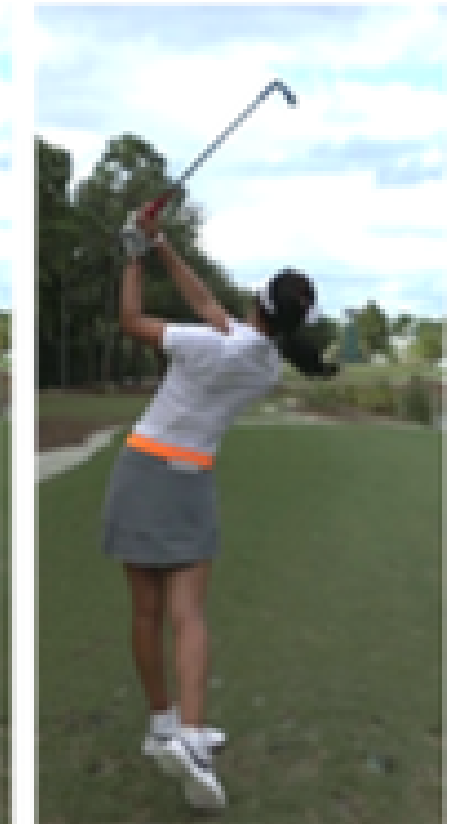
Mid-downswing



Impact



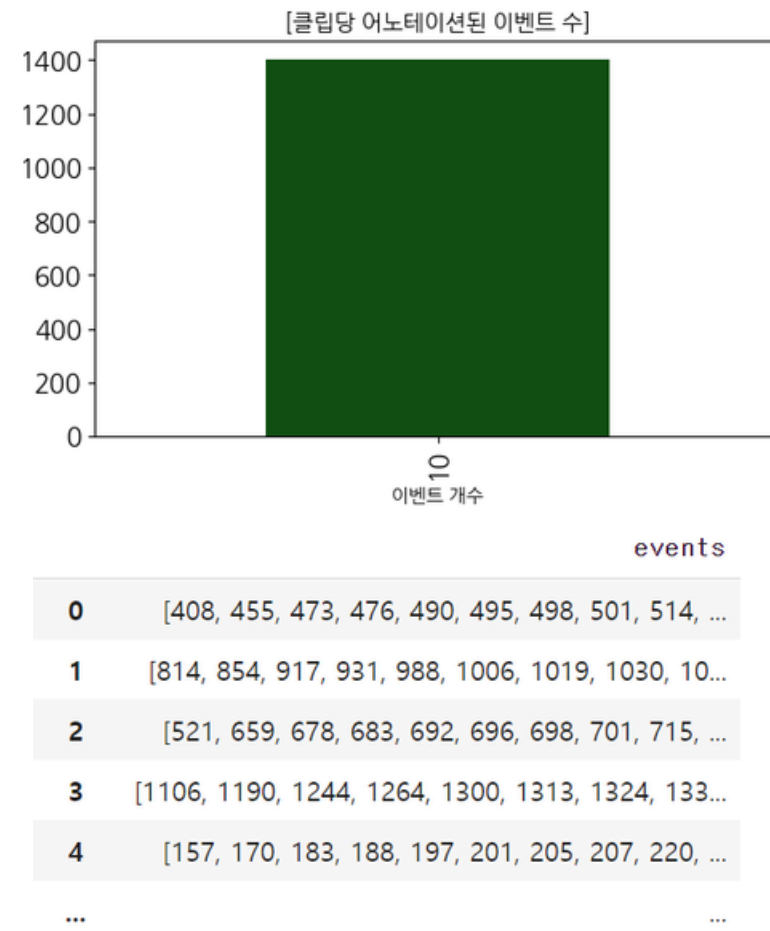
Mid-follow-through



Finish

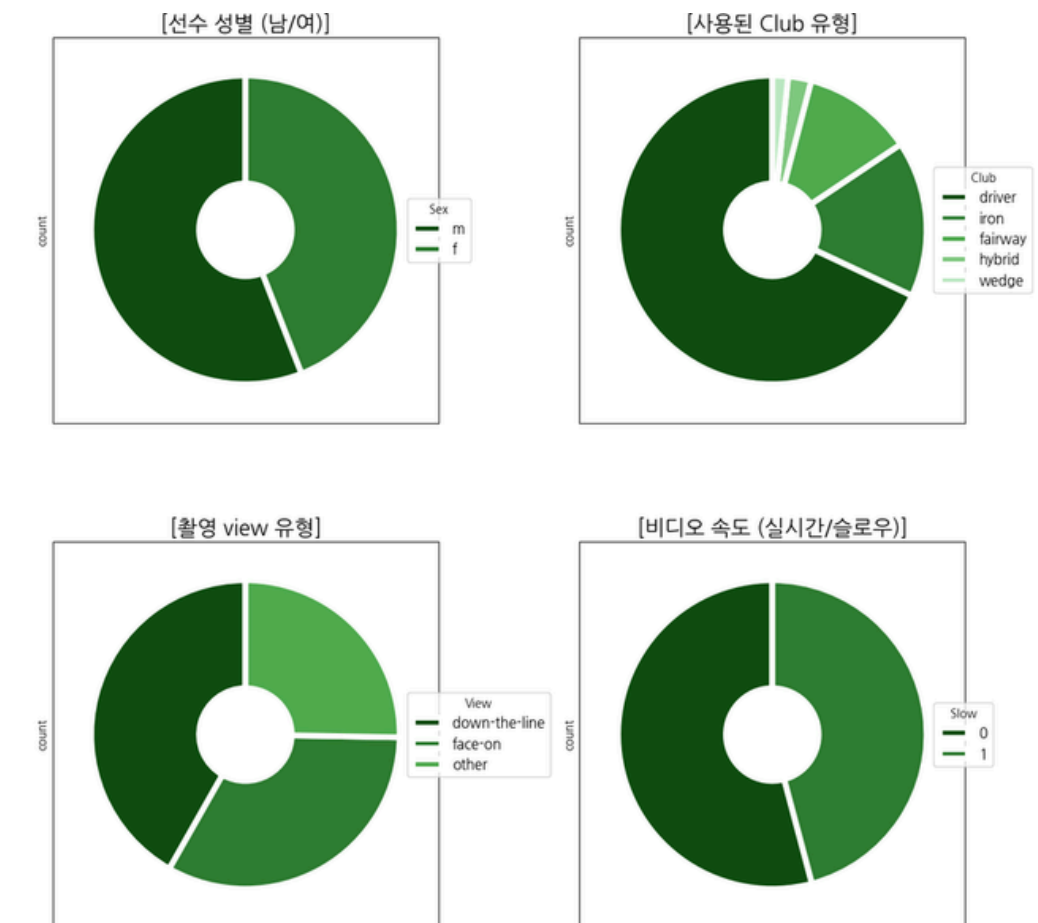
GolfDB EDA_1

1



- GolfDB: 총 580개 유튜브 영상에서 **1,400개 스윙 클립**을 추출함
각 클립마다 **10개의 이벤트가 어노테이션**되어 있음
→ 8개의 골프 스윙 이벤트 + 샘플의 시작, 끝 부분

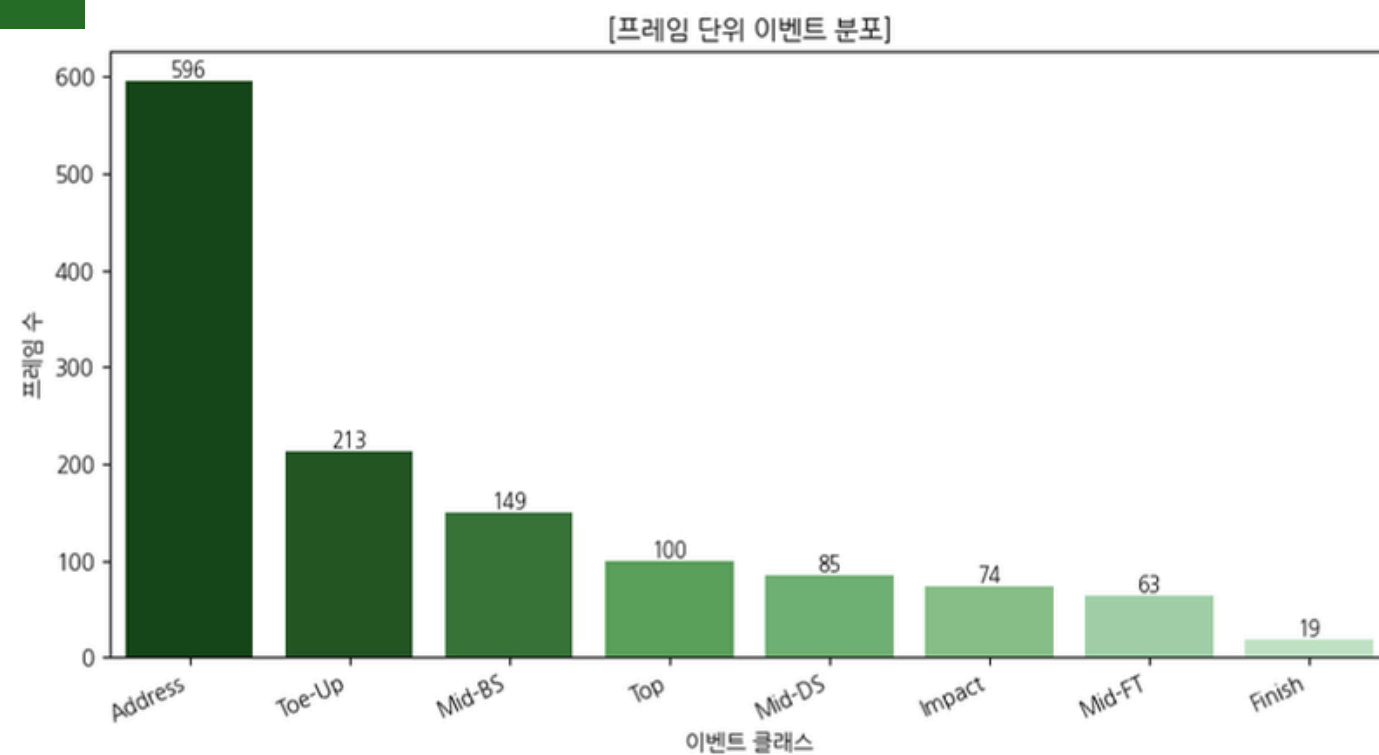
2



- 성별: 남 55 % | 여 45 % → 분포 균일
- Club: 드라이버가 68 %, 아이언 16 %
- View: 다양한 각도 확보, 단 Down-the-line 우세
- 속도: 실시간 54 % | 슬로우 46 % → 분포 균일

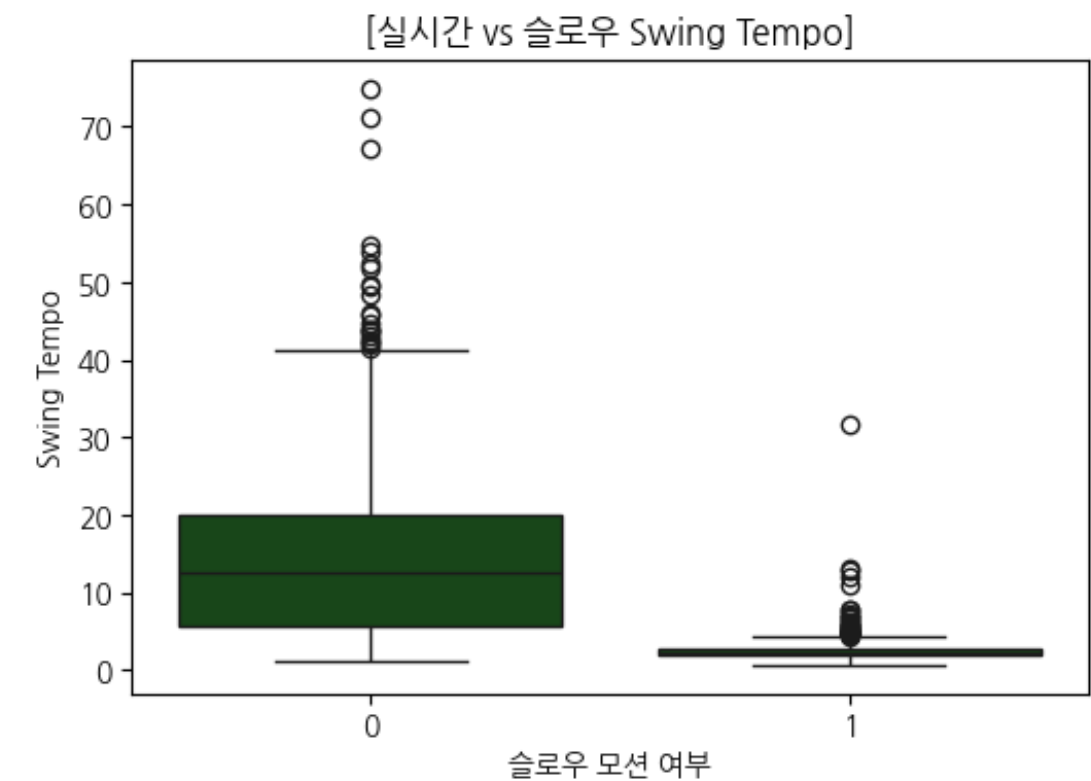
GolfDB EDA_2

3



- 초반(Address · Toe-Up) 프레임이 70 % 이상
⇒ 모델이 “스윙 시작 자세”에 과도하게 최적화될 가능성 ↑
- 후반(Impact 이후) 프레임은 합계 14 % 미만
⇒ Finish 인식 정확도 ↓, 클래스 가중치/오버샘플링 필요

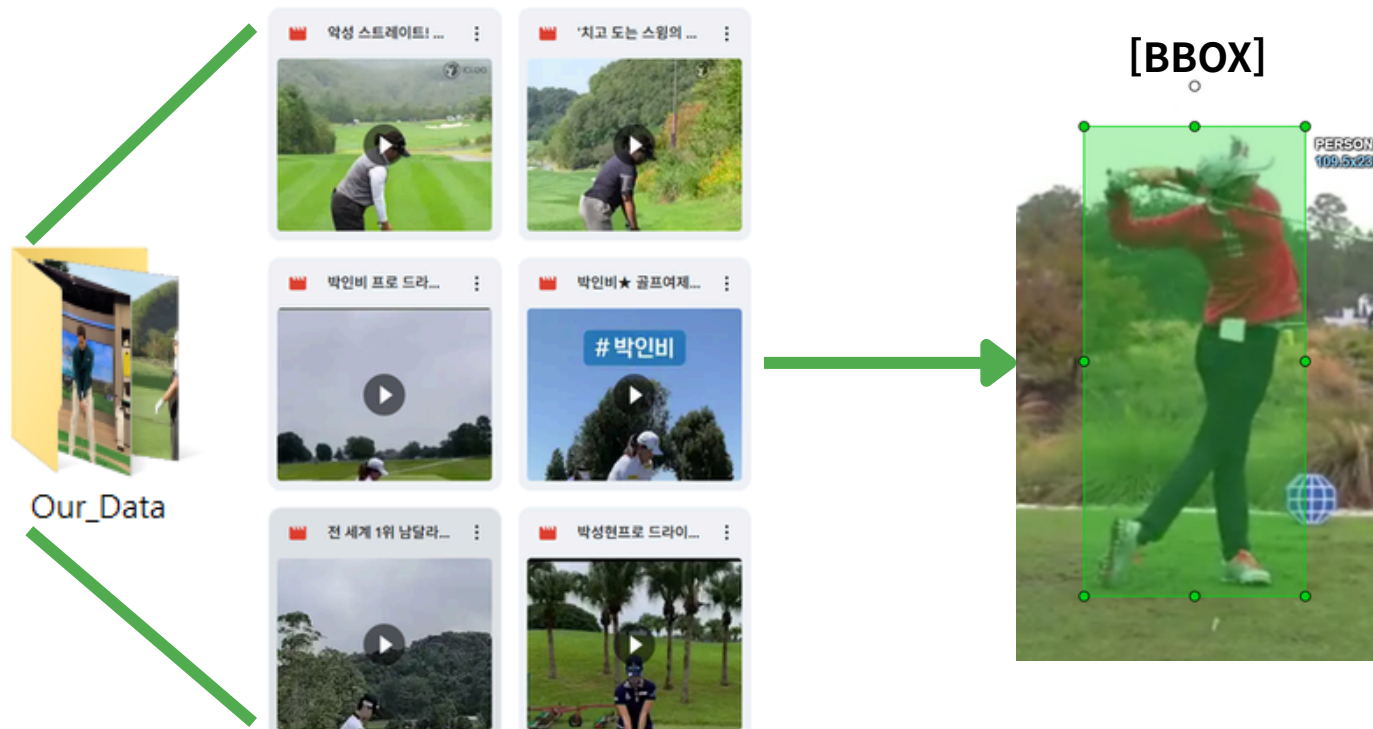
4



[swing_tempo = Back-swing 프레임 ÷ Down-swing 프레임]

- 중앙값: 실시간 12.5 F | 슬로우 2.3 F
⇒ 슬로우는 프레임 밀집도가 5배 높아 Feature Scaling 필요
⇒ Tempo 분포의 차이가 크므로 SwingNetV2 학습 시
fps 차이를 보정하지 않으면 과적합으로 인해 정확도 ↓

SwingNetV2_Annotation



id	youtube_id	events	bbox
0	10 xD6KDqPF9cc	[3, 10, 27, 30, 37, 41, 45, 47, 56, 60]	[0.1296875, 0.0006944444444444445, 0.71015625, ...]
1	11 XlzyJGDBuqc	[0, 8, 32, 35, 45, 50, 54, 56, 63, 72]	[0.07421875000000001, 0.0006944444444444445, 0....
2	12 XIAFMYH32ng	[6, 16, 28, 31, 37, 44, 48, 50, 58, 63]	[0.2125, 0.05, 0.6, 0.9208333333333333]
3	13 xS9NnBGgtjg	[2, 14, 23, 27, 32, 34, 38, 40, 47, 50]	[0.178125, 0.0006944444444444445, 0.484375, 0.]
4	14 Xx4vxPvunuk	[6, 18, 31, 39, 49, 53, 57, 60, 74, 80]	[0.11640625, 0.0006944444444444445, 0.53671875, ...]

[Youtube 영상 수집]

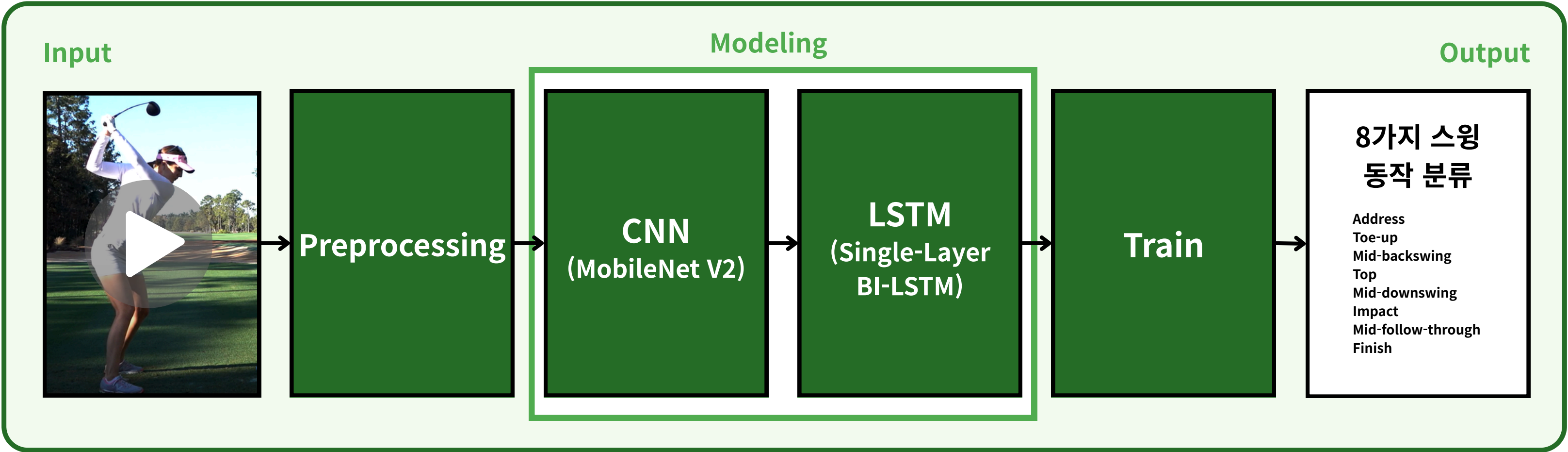
- 국내 남녀 프로 골퍼 및 아마추어 선수
- 다양한 실내외 환경에서 촬영한 영상
- 3분 미만 화면비 무관한 shorts 영상 위주 수집
- 스윙자세 전과정 원테이크 영상

[Annotation Tool : CVAT]

- 기준 자세 8가지 동일
- BBOX 라벨링 후 (0, 1) Normalization 적용
- 전처리를 위해 일정하게 인물 위치 crop

모델 개발

SwingNetV1



Config.	Input Size (<i>d</i>)	Seq. Length (<i>T</i>)	Batch Size	ImageNet Weights	LSTM Layers (<i>N</i>)	LSTM Hidden (<i>H</i>)	Bidirect.	Params (10 ⁶)	FLOPs (10 ⁹)	PCE (10k iter.)
0	224	32	6	Yes	2	128	Yes	4.07	10.32	66.8
1	224	32	6	No	2	128	Yes	4.07	10.32	1.5
2	224	32	6	Yes	2	128	No	3.08	10.26	54.7
3	192	32	6	Yes	2	128	Yes	4.07	7.62	45.7
4	160	32	6	Yes	2	128	Yes	4.07	5.33	62.4
5	128	32	6	Yes	2	128	Yes	4.07	3.45	57.7
6	160	64	6	Yes	2	128	Yes	4.07	10.65	71.1
7	160	32	12	Yes	2	128	Yes	4.07	5.33	70.1
8	224	32	6	Yes	1	128	Yes	3.67	10.39	69.4
9	224	32	6	Yes	2	64	Yes	3.01	10.26	66.9
10	224	32	6	Yes	2	256	Yes	6.96	10.51	69.3

SwingNetV1_문제점

1. 성능지표
2. CNN Backbone
3. LSTM Layer
4. Augmentation 및 전처리
5. 학습 테크닉
6. 단순 8개 동작 시퀀싱



SwingNetV1_Profiling

*GolfDB논문 성능지표

Seq. Length (T)	FLOPs (10 ⁹)	CPU (ms)*	PCE
64	10.92	10.6	76.2
32	5.41	10.8	74.0
16	2.70	11.5	71.0
8	1.35	12.0	66.0
4	0.68	13.8	63.1

Self CPU time total: 467.273ms
Self CUDA time total: 467.606ms

CPU

Name	Self CPU %	Self CPU	CPU total %	CPU total	CPU time avg	CUDA Mem	Self CUDA Mem	# of Calls
cudaEventRecord	61.94%	289.419ms	61.94%	289.419ms	125.235us	0 b	0 b	2311
cudaDeviceSynchronize	22.61%	105.635ms	22.61%	105.635ms	105.635ms	0 b	0 b	1
cudaLaunchKernel	6.51%	30.400ms	6.51%	30.400ms	60.878us	0 b	0 b	501
CNN_PASS	2.05%	9.563ms	74.82%	349.582ms	349.582ms	5.67 Gb	-15.97 Gb	1
aten::cudnn_batch_norm	0.83%	3.868ms	22.80%	106.524ms	2.049ms	9.75 Gb	0 b	52
aten::_convolution	0.80%	3.727ms	15.61%	72.961ms	1.403ms	9.75 Gb	0 b	52
aten::zeros	0.60%	2.817ms	0.62%	2.909ms	1.455ms	0 b	0 b	2
aten::hardtanh_	0.52%	2.423ms	11.68%	54.580ms	1.559ms	1.81 Gb	0 b	35

● CPU 비효율 개선 필요
[이벤트 기록 + 동기화 + 커널 런칭]
대기시간이 차지하는 비율 : 91%

● GPU 작동 중 병렬로 사용가능
전처리 과정 CPU 1~2%미만 소모

Self CPU time total: 467.273ms
Self CUDA time total: 467.606ms

GPU(CUDA)

Name	Self CPU %	Self CUDA	Self CUDA %	CUDA total	CUDA time avg	CUDA Mem	Self CUDA Mem	# of Calls
aten::cudnn_batch_norm	0.83%	180.239ms	38.55%	181.112ms	3.483ms	9.75 Gb	0 b	52
aten::cudnn_convolution	0.38%	91.710ms	19.61%	91.710ms	2.620ms	6.39 Gb	6.39 Gb	35
aten::_conv_depthwise2d	0.08%	82.690ms	17.68%	82.758ms	4.868ms	3.36 Gb	0 b	17
aten::clamp	0.13%	78.345ms	16.75%	78.415ms	2.240ms	0 b	0 b	35
aten::copy_	0.06%	16.042ms	3.43%	16.042ms	763.905us	0 b	0 b	21
aten::add	0.04%	4.116ms	0.88%	4.116ms	411.600us	323.44 Mb	323.44 Mb	10
aten::_convolution	0.80%	3.331ms	0.71%	177.799ms	3.419ms	9.75 Gb	0 b	52
aten::_cudnn_rnn	0.19%	2.865ms	0.61%	2.945ms	2.945ms	10.62 Mb	-59.59 Mb	1

● backbone의 효율/경량화 필요
상위 4개의 cnn연산이 GPU의 92%

● 레이어 추가시 정확도 향상 기대
LSTM의 소요 비중 1%미만

SwingNetV1 기반의 새로운 모델 SwingNetV2 제시



사용자의 스윙 영상을
시퀀스별로 분류하여 골프 스윙의 객관적 지표를
구현한 SwingNet 기반의 시스템

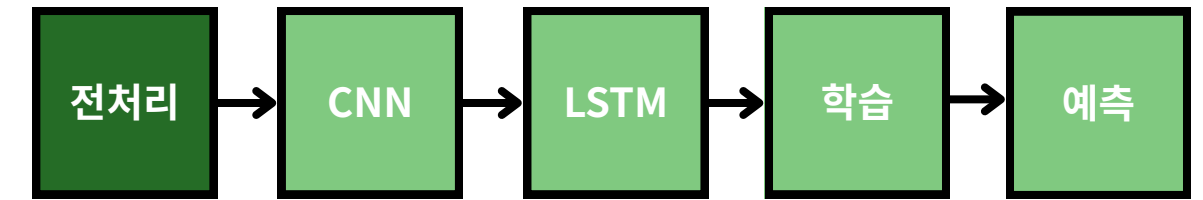


Augmentation

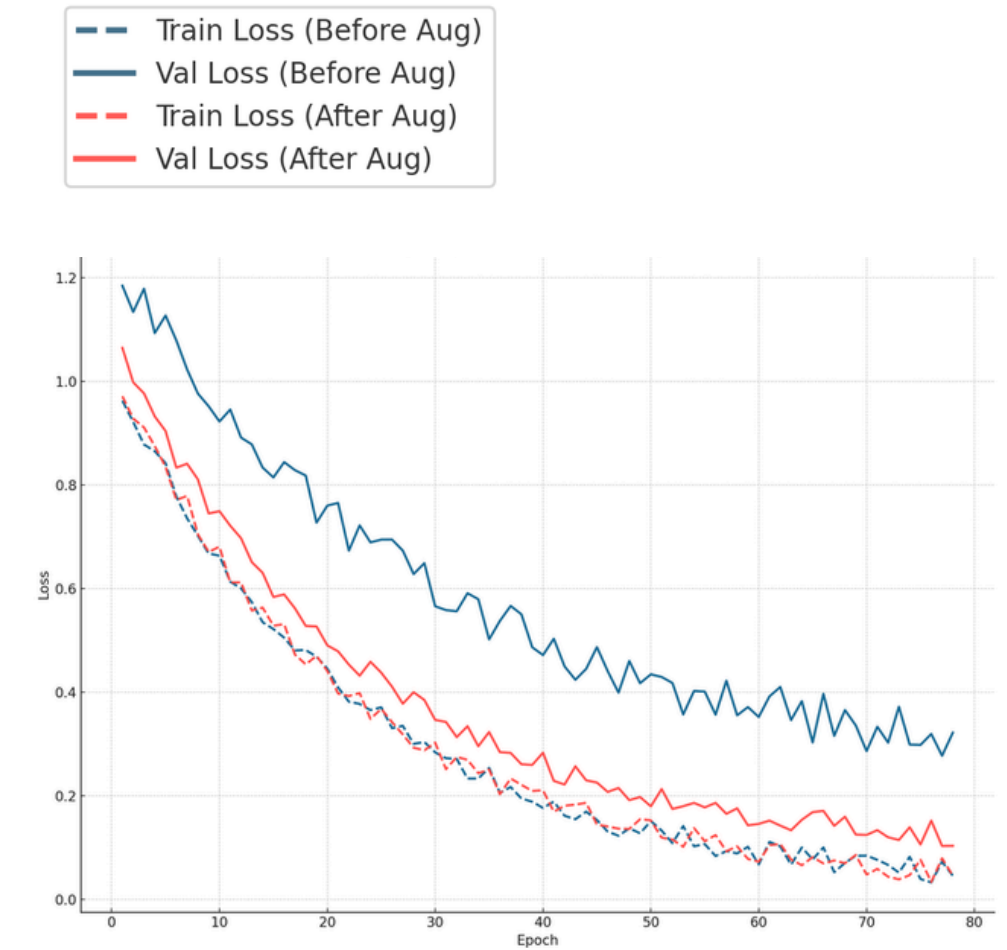
시도



- Random Horizontal Flip
왼손/오른손잡이 고려하여 $p=0.5$ 증강
- Random Rotate
 $-5^\circ \sim 5^\circ$ 카메라 각도 불균형 보정

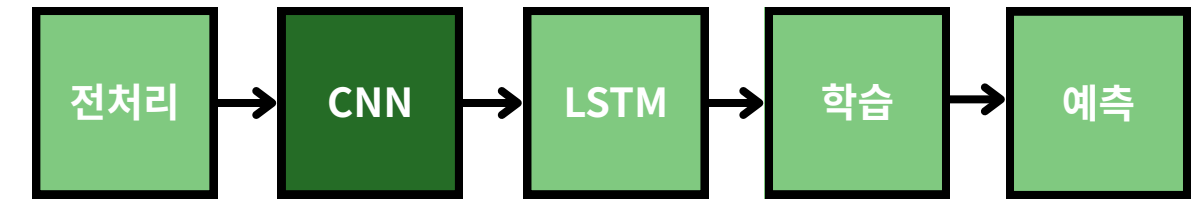


결과

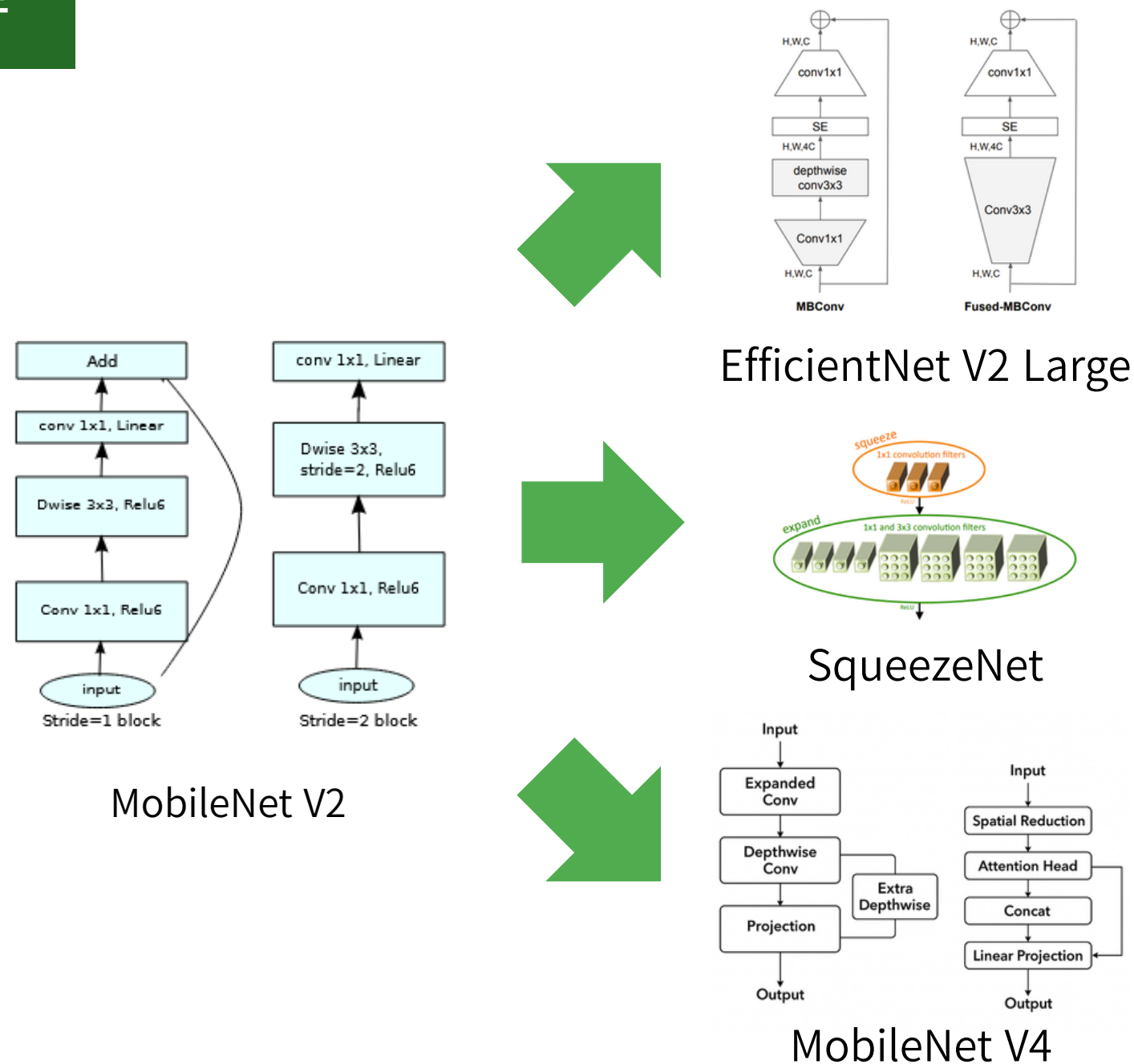


Before/After Graph

CNN 변경



시도



결과

Backbone 성능 비교

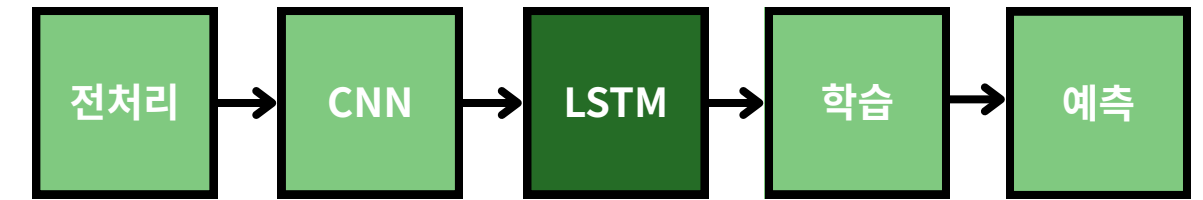
모델	Params (M)	FLOPs (G)	Latency (ms)	Top-1 Acc.	PCE (%)
MobileNetV2 (2017)	3.4	0.6	4	71.88%	76.1
SqueezeNet (2016)	1.2	0.72	4.8	58.18%	74.5(E)
Efficient Net V2-B0 (2021)	118.5	56.08	378	85.81%	80.6(E)
Mobilenet V4 (2024)	3.8	0.2	1.3	73.80%	77.8(E)

실시간성: **MobileNetV4** (1.3 ms) > MobileNetV2(4ms) > SqueezeNet (~4.8 ms)

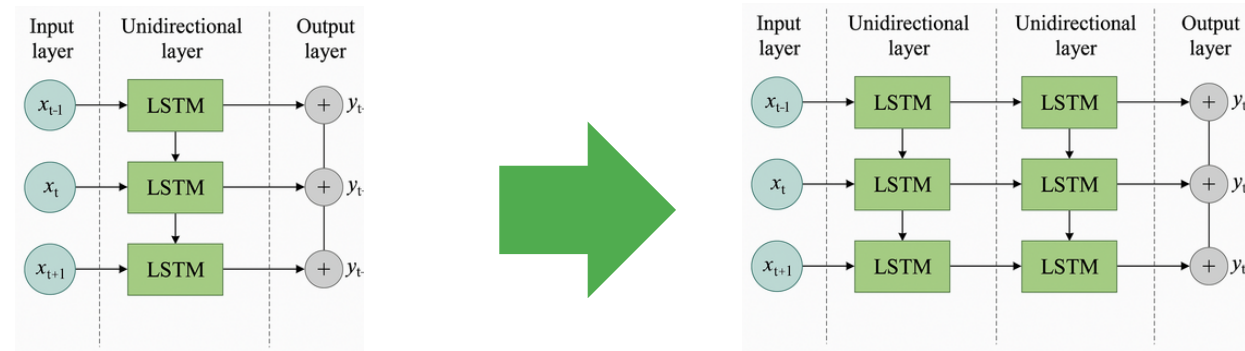
정확도: EfficientNet V2 Large > **MobileNetV4** > MobileNetV2

적은 데이터 환경에서 Accuracy 대비 Latency 성능
“MobileNetV4”가 가장 우수

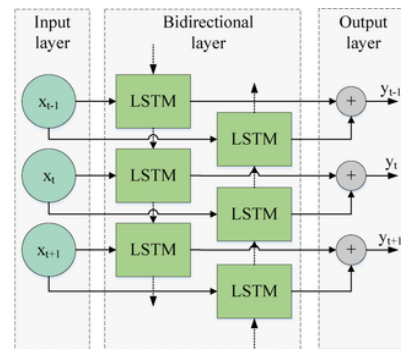
LSTM 변경



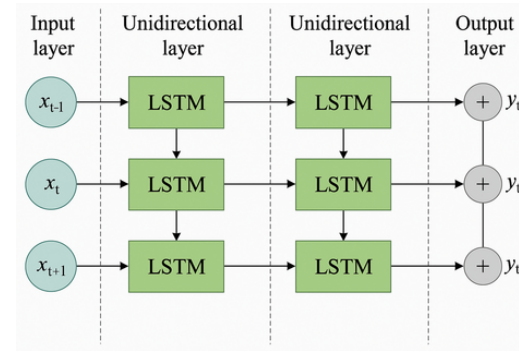
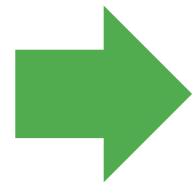
시도



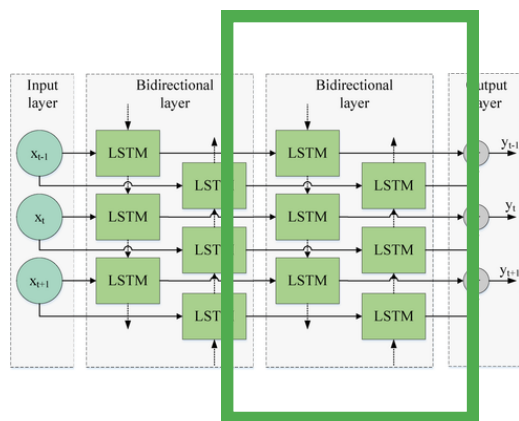
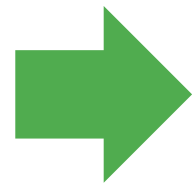
1 Layer
Uni-directional



1 Layer
Bi-directional

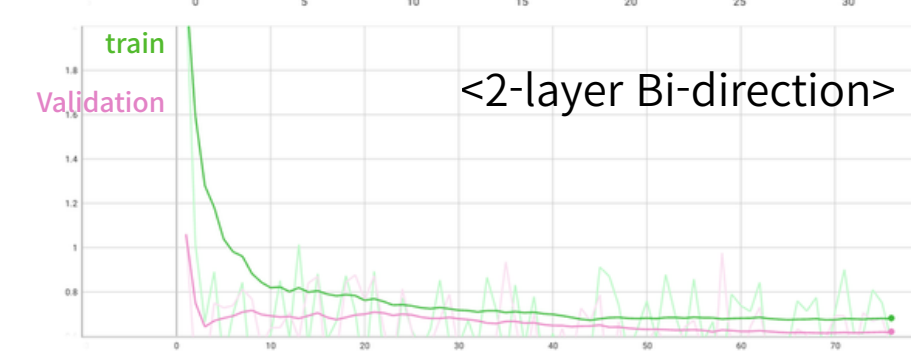
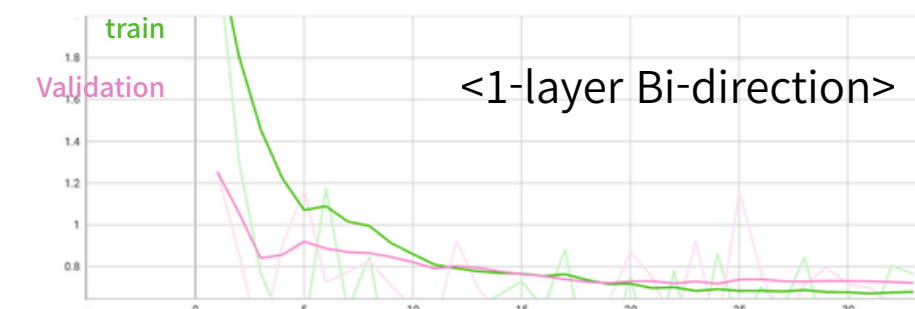
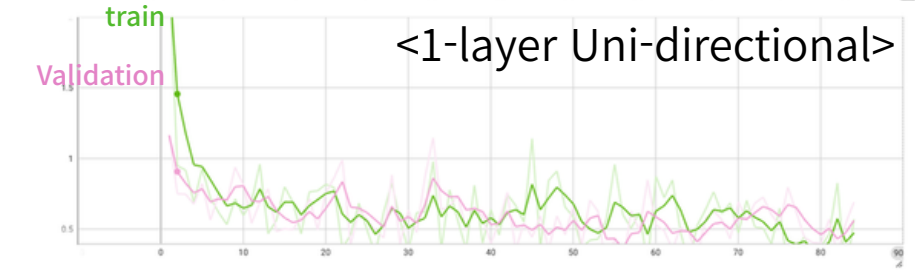
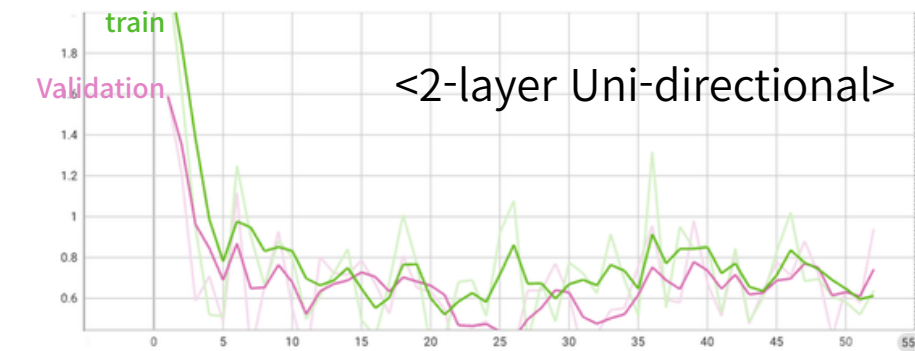


2 Layer
Uni-directional

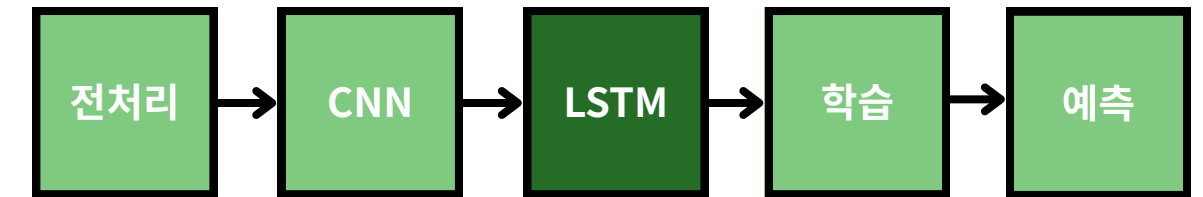


2 Layer
Bi-directional

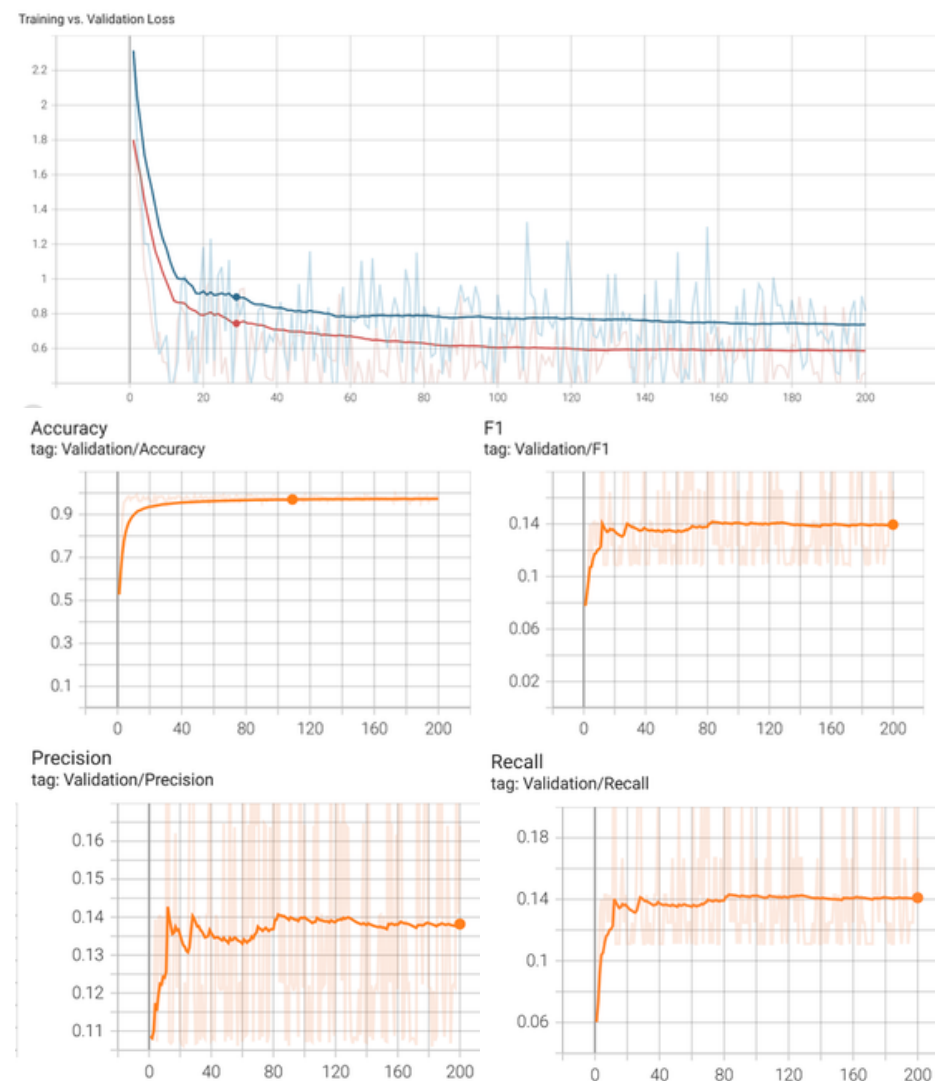
결과



ConVLSTM



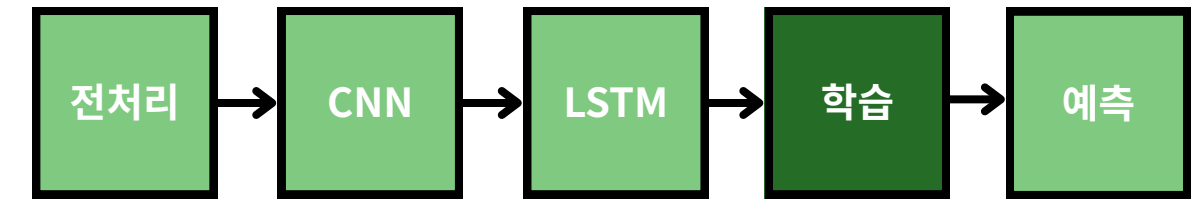
시도



결과

- Training vs Validation Loss
에폭 0-20: 손실 급락 → 이후 0.5 수준에서 안정화
Training/Validation 차이 작아 과적합 적음
- 정확도(Accuracy)
에폭 20 이전에 0.90 도달 → 최종 0.92 수준 유지
- F1-Score 및 Precision
초기 0.11 → 0.14 상승 후 정체 → 클래스 불균형
- Recall
초기 0.06 → 0.14로 개선 후 0.14 수준 머무름
→ 불균형 영향 지속

학습 Hyper-Parameter Tunning



시도

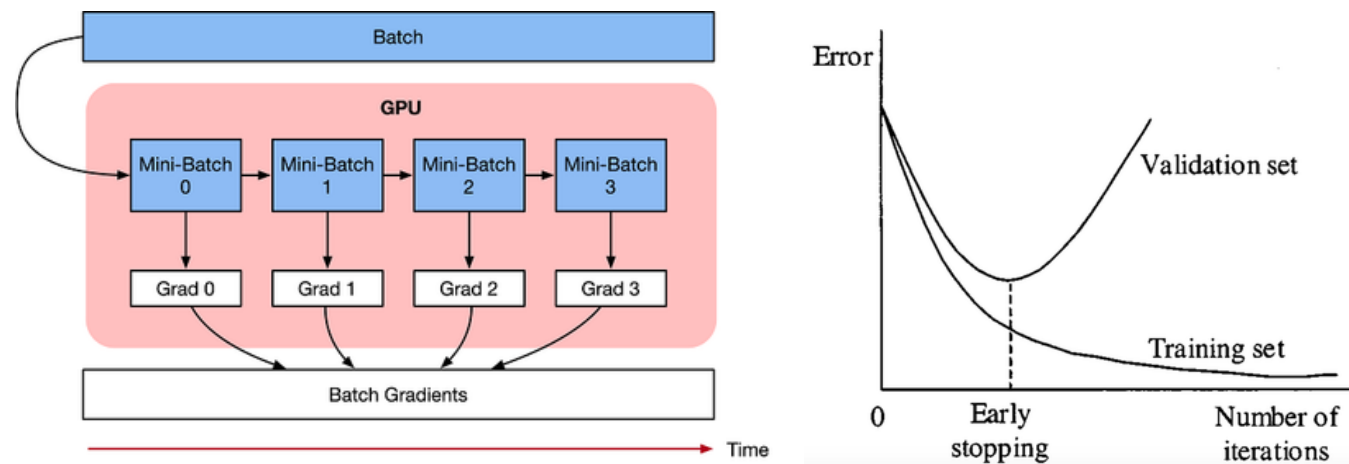
- # bs = batch size
- # early-stopping 적용
- # gradient accumulation 적용

Loss Function

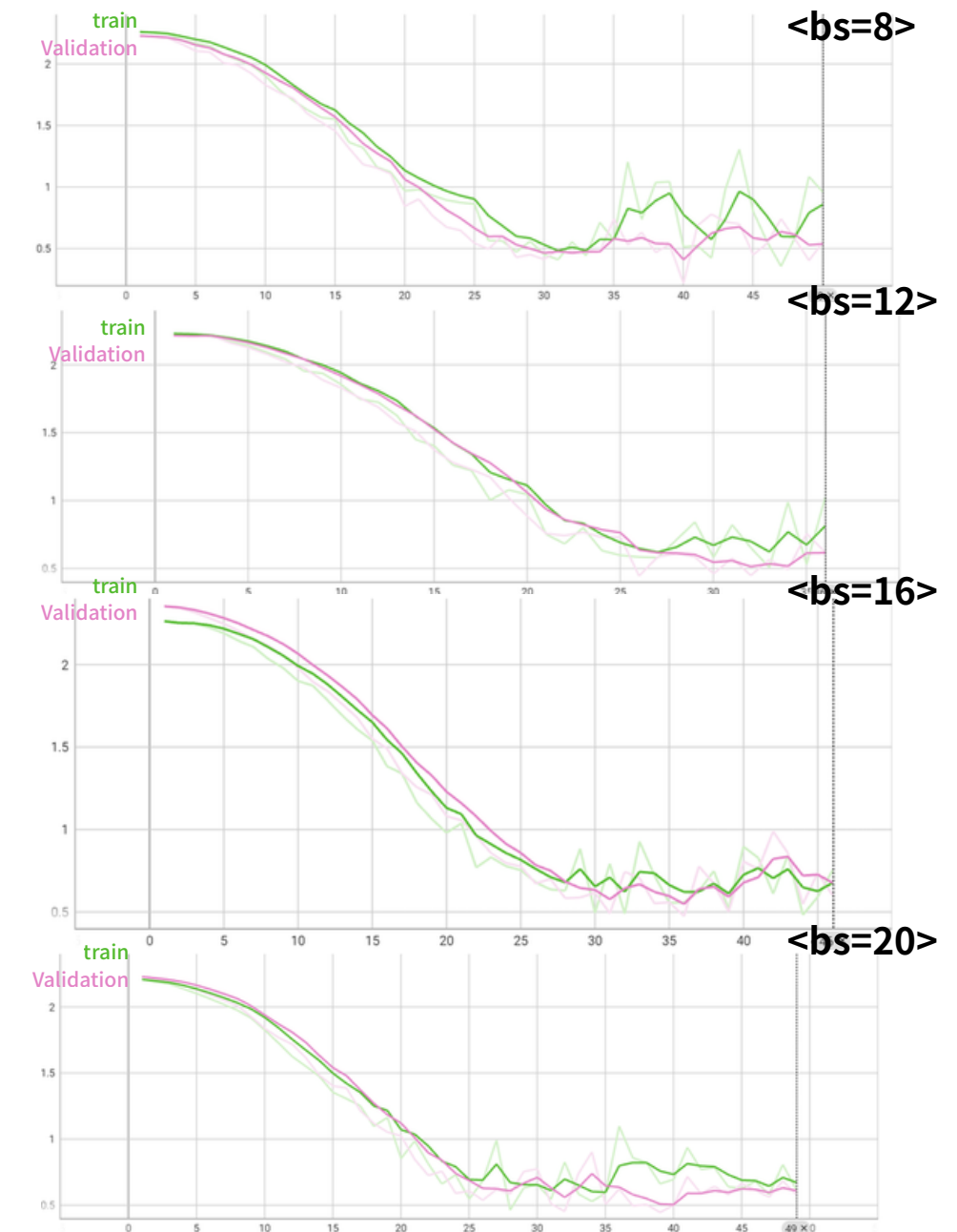
$$Loss(\theta) = \frac{1}{2} (h(x^k) - y^k)^2$$

Gradient and weight function

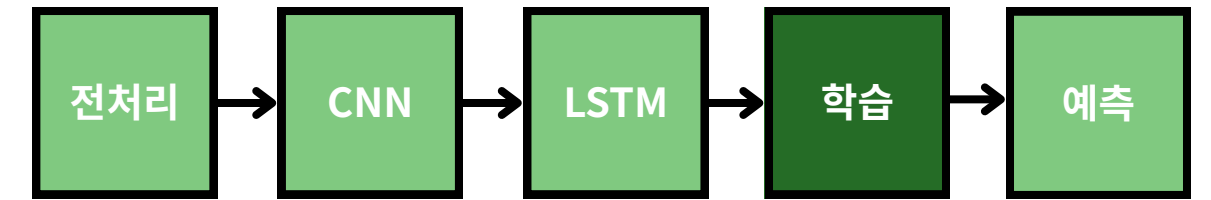
$$accumulated = \sum_{i=0}^N grad_i \quad \theta_i = \theta_{i-1} - lr * \sum_{i=0}^N grad_i$$



결과



Learning Rate Scheduler / Warm up



시도

LAMBDA LR

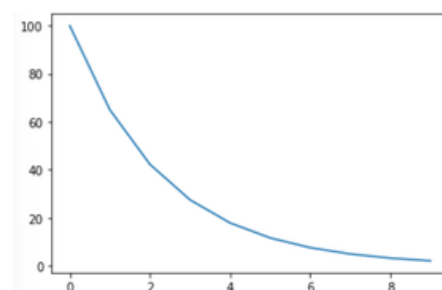
$$lr_{epoch} = lr_{initial} * \text{Lambda}(\text{epoch}) \quad | \text{reepoch} = \text{lr}_{initial} * \text{Lambda}(\text{epoch})$$

ExponentialLR

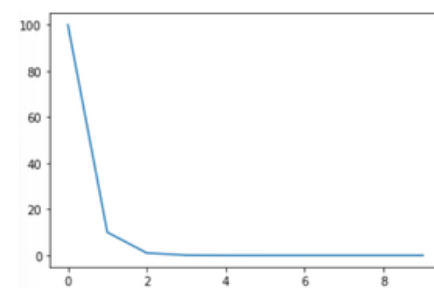
$$lr_{epoch} = \text{Gamma} * lr_{epoch-1} \quad | \text{reepoch} = \text{Gamma} * \text{lr}_{epoch-1}$$

CosineAnnealingLR

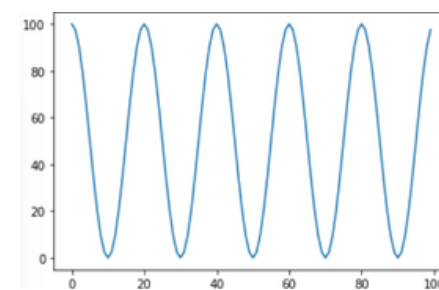
$$\eta_t = \eta_{min} + \frac{1}{2}(\eta_{max} - \eta_{min}) \left(1 + \cos \left(\frac{T_{cur}}{T_{max}} \pi \right) \right)$$



LAMBDA LR

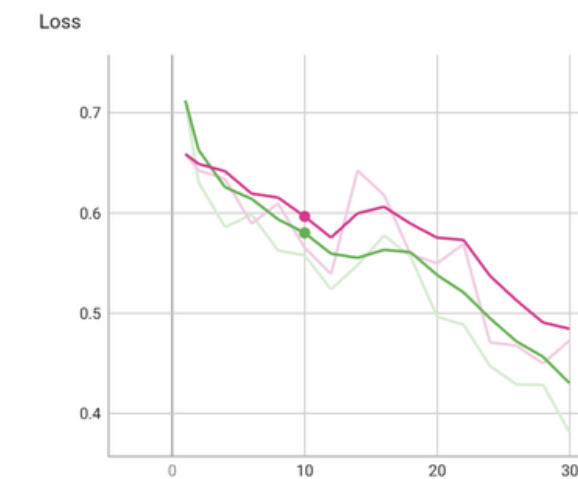


ExponentialLR



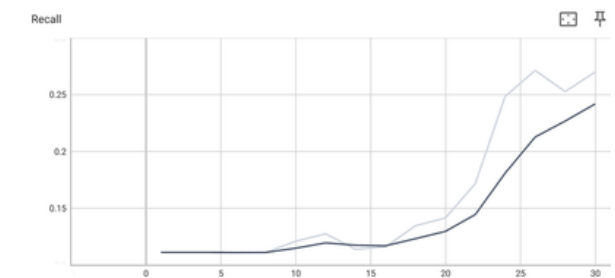
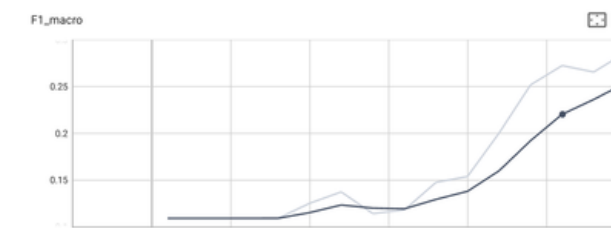
CosineAnnealingLR

결과

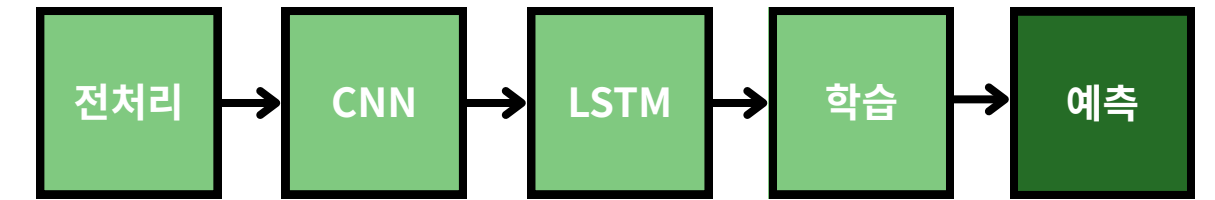


Warm-up: 초기 안정화를 달성

Scheduler: 후반 성능 극대화



가이드 활용 지표



세 점 사이의 각도 계산 - angle

$$r = \text{atan2}(y_C - y_B, x_C - x_B) - \text{atan2}(y_A - y_B, x_A - x_B),$$

$$\alpha = \left\lfloor \frac{180}{\pi} r \right\rfloor,$$

$$\text{angle} = \begin{cases} 360 - \alpha, & \alpha > 180, \\ \alpha, & \alpha \leq 180. \end{cases}$$

두 점 사이의 기울기 계산 - inclination

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2},$$

$$d \cdot x_1 = 0 \implies \boxed{\text{angle} = 0},$$

$$\cos \theta = \begin{cases} \frac{(x_2 - x_1)(-x_1)}{d x_1}, & \text{axis} = x, \\ \frac{(y_2 - y_1)(-y_1)}{d y_1}, & \text{axis} = y, \end{cases} \quad K = \text{int}\left(\frac{180}{\pi}\right),$$

$$\theta = \arccos(\cos \theta),$$

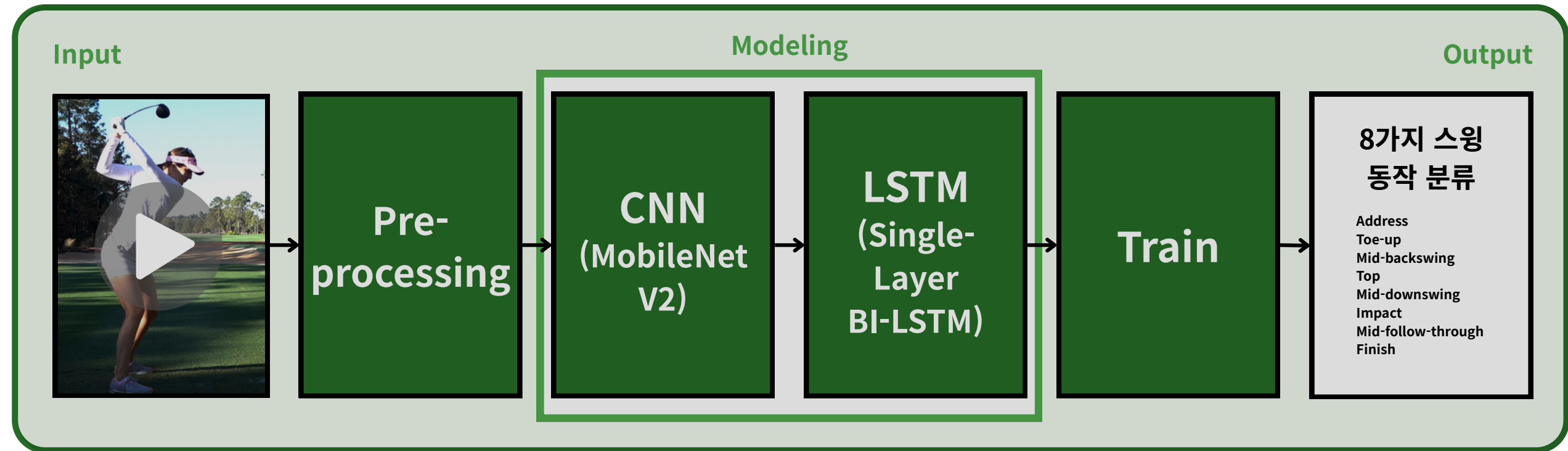
$$\text{angle} = \begin{cases} K \theta, & \text{orientation} = \text{right}, \\ 180 - K \theta, & \text{orientation} = \text{left}. \end{cases}$$



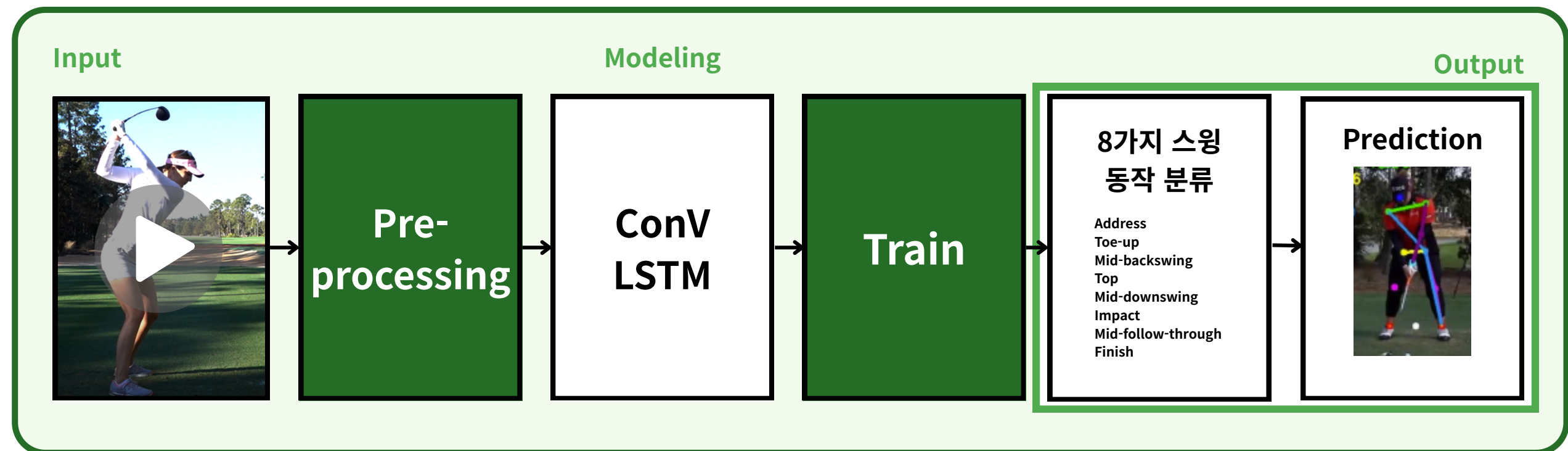
point : 13
angle : 3
inclination : 2

SwingNetV1, V2 비교

SwingNetV1



SwingNetV2



검증 및 테스트

검증 및 테스트

input video : <https://www.youtube.com/watch?v=3wru0WH0buk>



Address



Toe-up



Mid-backswing



Top



Mid-downswing



Impact



Mid-follow-through

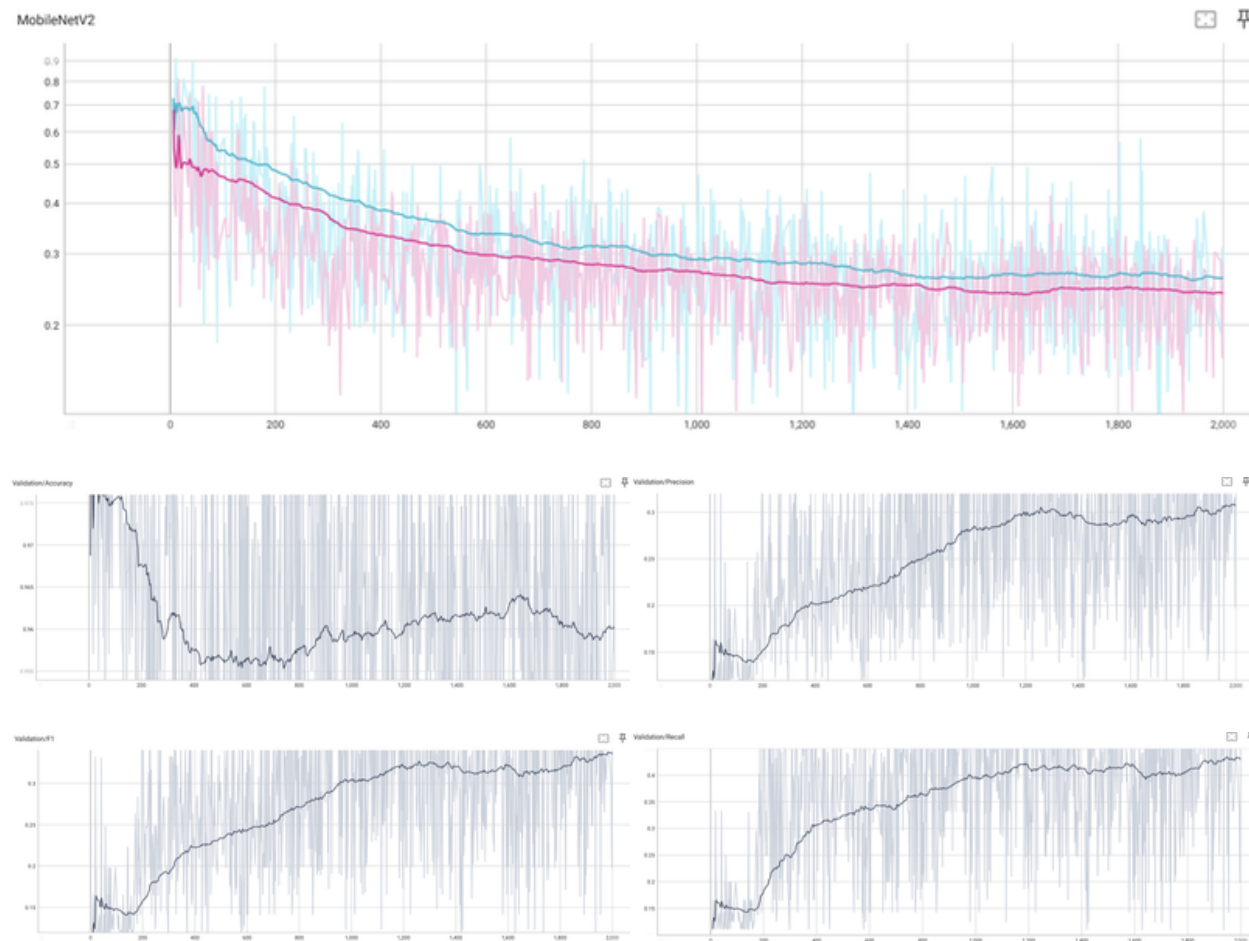


Finish

검증 및 테스트_2

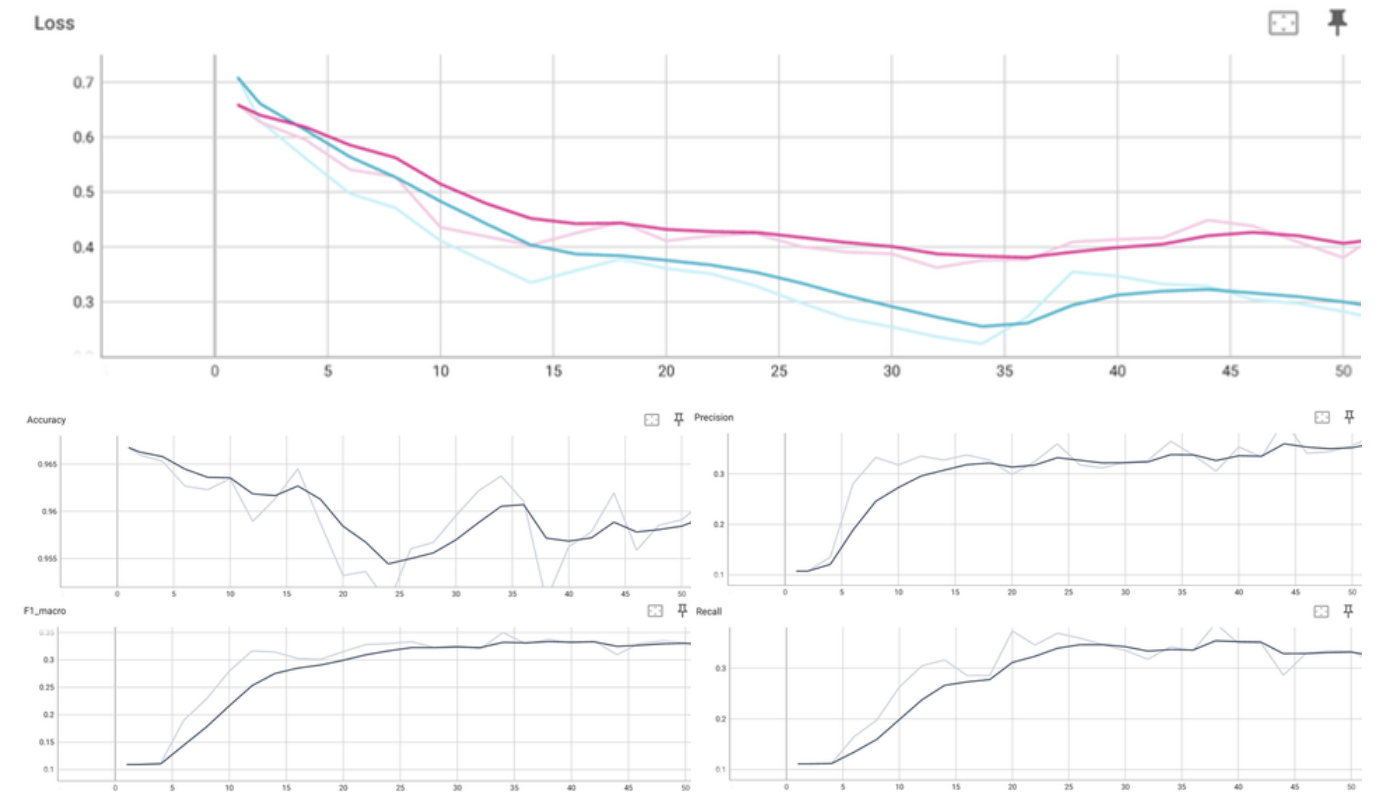
V1

SwingNetV1



V2

SwingNetV2



감사합니다

1조 서상원 안지현 임연오
